

Corso di Laurea [Magistrale?] in Ingegneria e Scienze Informatiche

Fancy Title

Tesi di laurea in:
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Abstract

Max 2000 characters, strict.

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Introduzione

Inquadramento del problema

Il problema dello School Timetabling

Obiettivi della tesi

Struttura della Tesi

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Chapter 1

Il Problema dell'HSTP e il Modello di Colorazione di Grafi

1.1 Definizione formale del High School Timetabling Problem (HSTP)

Definiamo gli insiemi fondamentali del sistema:

- Insieme dei Docenti: $D = \{d_1, d_2, \dots, d_{|D|}\}$
- Insieme delle Classi: $C = \{c_1, c_2, \dots, c_{|C|}\}$
- Insieme delle Aule/Laboratori: $A = \{a_1, a_2, \dots, a_{|A|}\}$
- Insieme degli Slot Temporal: $T = \{t_1, t_2, \dots, t_{|T|}\}$

Lo spazio delle soluzioni può essere modellato mediante variabili decisionali booleane:

$$x_{d,c,a,t} = \begin{cases} 1 & \text{se il docente } d \text{ insegna alla classe } c \text{ nell'aula } a \text{ nello slot } t \\ 0 & \text{altrimenti} \end{cases}$$

dove $d \in D$, $c \in C$, $a \in A$, $t \in T$.

1.2 Astrazione del problema: Riduzione al Graph Coloring Problem (GCP)

1.2.1 Costruzione del Grafo di Conflitto $G = (V, E)$

Definiamo il grafo di conflitto $G = (V, E)$, dove:

- L'insieme dei nodi V rappresenta gli eventi elementari (es. singola ora di lezione o incontro didattico docente-classe).
- L'insieme degli archi E rappresenta i conflitti (archi tra nodi che non possono essere programmati contemporaneamente, ad esempio perché condividono lo stesso docente o la stessa classe).

1.2.2 Mappatura degli slot temporali

Sia K l'insieme dei colori disponibili. Associamo in modo biunivoco ogni colore $k \in K$ a uno slot temporale $t \in T$, con la condizione che $|K| = |T|$. Il problema di scheduling si riduce alla ricerca di una colorazione ammissibile dei nodi di G .

1.2.3 Estensioni del modello standard

1.3 Analisi della complessità computazionale

Chapter 2

Traduzione Formale e Matematicizzazione dei Vincoli d'Istituto (A.S. 2026/2027)

In questo capitolo viene presentata la formulazione matematica rigorosa dei vincoli che governano l'orario scolastico dell'Istituto Tecnico Saffi-Alberti di Forlì per l'anno scolastico 2026/2027.

2.1 Formalizzazione degli Hard Constraints (Vincoli di Ammissibilità Strutturale)

I vincoli rigidi (hard constraints) devono essere necessariamente soddisfatti affinché l'orario sia considerato valido (ammissibile).

2.1.1 Vincoli di unicità ed esclusione

Ogni docente e ogni classe non possono svolgere più di una lezione contemporaneamente in un dato slot temporale.

$$\sum_{c \in C} \sum_{a \in A} x_{d,c,a,t} \leq 1 \quad \forall d \in D, \forall t \in T \quad (2.1)$$

$$\sum_{d \in D} \sum_{a \in A} x_{d,c,a,t} \leq 1 \quad \forall c \in C, \forall t \in T \quad (2.2)$$

2.1.2 Limitazione delle risorse fisiche

Un'aula (o laboratorio/palestra) $a \in A$ non può ospitare più di una classe (o un evento di lezione) contemporaneamente nello slot temporale $t \in T$.

$$\sum_{d \in D} \sum_{c \in C} x_{d,c,a,t} \leq 1 \quad \forall a \in A, \forall t \in T \quad (2.3)$$

2.1.3 Sincronizzazione delle compresenze

2.1.4 Condizioni al contorno esterne

$$x_{d,c,a,t} = 0 \quad \forall (d, c, a, t) \text{ tali che lo slot } t \text{ rientra nell'indisponibilità} \quad (2.4)$$

2.2 Formalizzazione dei Soft Constraints e Costruzione della Funzione Obiettivo

I vincoli morbidi (soft constraints) descrivono preferenze e criteri di qualità didattica e organizzativa. L'obiettivo è minimizzare le violazioni complessive.

2.2.1 Funzione di costo complessiva

Definiamo $P_i(x)$ come la funzione che misura la violazione del vincolo soft i -esimo, e $w_i \geq 0$ il peso associato a tale vincolo. La funzione obiettivo complessiva è espressa come:

$$\min z = \sum_i w_i \cdot P_i(x) \quad (2.5)$$

2.2.2 Densità didattica ed equilibrata ripartizione

2.2.3 Vincoli di sbarramento di adiacenza

2.2.4 Equità del carico di lavoro docente

2.2.5 Vincoli di contiguità e comfort

2.2.6 Logistica multi-plesso e partizione temporale

2.2.7 Rilassamento dei vincoli e priorità

Chapter 3

Paradigmi Algoritmici di Risoluzione per l'HSTP

In questo capitolo vengono esaminati i principali paradigmi algoritmici utilizzati in letteratura e nella pratica ingegneristica per risolvere il problema dell'High School Timetabling Problem (HSTP).

3.1 Metodi Esatti: Programmazione Lineare Intera (ILP) e SAT-Solving

3.2 Constraint Programming (CP)

3.3 Metaeuristiche basate su Traiettorie (Ricerca Locale)

3.3.1 Tabu Search (TS)

3.3.2 Simulated Annealing (SA)

3.4 Metaeuristiche Evolutive (Algoritmi Genetici - GA)

Chapter 4

Analisi Comparativa e Discussione dell'Approccio Ottimo

4.1 Valutazione quantitativa dei paradigmi analizzati

Di seguito viene proposta una struttura tabellare per il confronto quantitativo e qualitativo dei diversi approcci analizzati:

Algoritmo / Paradigma	Tempo di Calcolo	Complessità Spaziale	Ottimalità	Gestione Complessità (Saffi-A)
ILP / SAT-Solving				
Constraint Programming				
Tabu Search				
Simulated Annealing				
Algoritmi Genetici				

Table 4.1: Confronto qualitativo e quantitativo delle metodologie risolutive.
ab:confronto-metodologie

4.2 Selezione dell'approccio algoritmico e dell'architettura software ideale

Conclusioni

Sintesi del lavoro svolto

Sviluppi futuri

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